JavaScript is not running. Open this file in Adobe Acrobat Reader on a computer, with JavaScript enabled.

Search a test, topic, or R function…

24 operations

Universal detector
Paste first, identify automatically

OPERATIONS

- Plotting & tables
- Summarize for stats
- Skewness & kurtosis
- Wide table format
- Confidence / prediction intervals
- One-sample mean / proportion
- Two-sample mean / variance
- >2 samples: ANOVA
- One-variable linear regression
- Multi-variable linear regression
- Model significance
- Pairwise model comparison
- Dummy variables / reference groups
- Interaction terms
- GLM / binary variable
- Confusion matrix
- PCA & component significance
- Forward/Backward Stepwise Selection
- K-means clustering
- K-fold cross-validation
- lpSolve / Sensitivity analysis
- Shapiro–Wilk
- Pearson correlation test

Universal detector

RStudio input

Paste your RStudio console output here…

Extra or missing spaces (up to 3) are supported.

Prose output Answer option

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

Untitled document
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Plotting & tables

Plot setup

Use sample

I would like to plot a … of

column2, column3, column4 against

column1 , titled

Mean bar plot of Column2, Column3, . The dataset is called

(default is "dataset").

Axis transformations

X-axis transformation

Y-axis transformation

Generated R code

Answer option

Copy code

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Calculate axis intervals

Off: basic plotting parameters and automatic axes. Applies to all plots.

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Summarize for stats

Columns to summarize

Data location (data-frame name)

datafile

Enter a column name. A new field appears automatically as you type.

column1 ×
column2 ×
column3 ×
column4 ×
column5 ×
column6 ×
column7 ×
column8 ×

Statistics to include

- Non-missing count (n)
- Mean
- Standard deviation
- Median
- Skewness
- Excess kurtosis
- Minimum
- Maximum
- Total rows
- Missing count
- Variance
- First quartile (Q1)
- Third quartile (Q3)
- Interquartile range
- Sum

Total rows includes missing values; n counts non-missing observations. Kurtosis is excess kurtosis.

Generated R code

Answer option

Copy code

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Skewness & kurtosis

Distribution coefficients Use sample

Coefficient of skewness

e.g. 0.750

Tail-weight measure Kurtosis | Excess kurtosis

e.g. 2.40

Prose output Answer option

Copy prose

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Pearson correlation test

Wide table format

Table mapping Use sample

Dataset name	dataset
Rows	column1
Column headers	column2
Values to summarize	column3
Table caption	Descriptive Statistics Table

Generated R code Answer option

Copy code

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Confidence / prediction intervals

Interval setup

Use sample

I want to calculate [ ] for [ ]

with [ ]

Dataset name

Column / parameter name
column1

Confidence level (%)

Sample mean (optional)
50

Sample standard deviation (optional)
4

Sample size (optional)
64

Calculated result

Copy prose | Answer option [ ] | Copy code

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

Interpret RStudio interval output

Use sample output | Answer option [ ] | Copy prose

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

RStudio output

```
upper_limit lower_limit
[1,] 20.33 19.32
```

Generated interpretation

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One-sample mean / proportion

Test setup

Use sample

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

I want to test for

Dataset name
dataset

Response column
column1

Hypothesized value
50

Significance level

Population filter (optional)
Product == "Book"

Exclude missing response values

Answer option

Copy code

RStudio output

Use sample output

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Paste the t.test() or prop.test() output here…

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Two-sample mean / variance

Independent two-sample t-test

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I want to test

Paired samples

Dataset name

Column 1 (response)

Column 2 (grouping)

Alternative hypothesis

Answer option

Copy code

RStudio output

Use sample output

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

Paste the two-sample test output here…

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>2 samples: ANOVA

Step 0: Setup

Use sample

Dataset name

Response variable

Grouping variable

Significance level

Normality can be assumed, or the dataset is large enough to invoke the Central Limit Theorem

Step 1: Test normality per group

Answer option

Copy code

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

Step 2: Test homogeneity of variance

Use sample output

Answer option

Copy code

Copy prose

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Number of groups

Generated R code

RStudio output

Paste the Bartlett or F-test output here…

Generated interpretation

Step 3a: Standard ANOVA + Tukey HSD

Use sample output

Answer option

Copy code

Copy prose

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

Generated R code

ANOVA and Tukey HSD output

Paste summary(model) and TukeyHSD(model) output here…

Generated interpretation

Step 3b: Welch ANOVA + Games–Howell

Use sample outputs

Answer option

Copy code

Copy prose

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

Generated R code

Welch ANOVA output

Paste the welch_anova_test() output here…

Games–Howell output

Paste the games_howell_test() pairwise table here…

Generated interpretation

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Shapiro–Wilk
Pearson correlation test

One-variable linear regression

Model setup

Use sample

Dataset name

Dependent variable (Y)

column2

Independent variable (X)

column1

Transform X

Generated R code

Answer option

Copy code

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Calculate axis intervals

Off: basic plotting parameters and automatic axes. Applies to all plots.

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- Pearson correlation test

Multi-variable linear regression

Multi-variable linear regression

Use sample

Dataset name

Dependent variable (Y)

Predictors, separated by commas

Model summary R code

Answer option

Copy code

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Prediction R code

Copy code

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Confidence interval R code

Copy code

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

Interpret regression output

Use sample output | Answer option | Copy all prose

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RStudio output

Paste the Call, Coefficients, and model summary from summary(lm(...)) here…

Model and predictor interpretations

Multiple R-squared Copy

Overall model F-test Copy

age Copy

educ Copy

income Copy

totwrk Copy

Coefficient confidence intervals

Use sample output | Answer option | Copy all prose

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

RStudio output

Paste the confidence-interval table from confint(linreg) here…

Parameter interpretations

totwrk Copy

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Search a test, topic, or R function…

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Model significance

Statistical significance of model

Use sample

Dataset name

Dependent variable (Y)

Predictors, separated by commas

Generated R code

Answer option

Copy code

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Pairwise model comparison

Pairwise model comparison

Use sample

Dataset name

Dependent variable (Y)

Restricted-model predictors

Full-model predictors

Generated R code

Answer option

Copy code

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Model-comparison interpretation

Use sample output

Answer option

Copy prose

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RStudio output

Paste the Analysis of Variance Table, including Model 1 and Model 2, here…

Generated interpretation

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Dummy variables / reference groups

Dummy variables / reference groups

Use sample

Dataset name

Dependent variable (Y)

Categorical variable

Levels, separated by commas

Reference level

Other predictors

Generated R code

Answer option

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Interaction terms

Interaction terms

Use sample

Dataset name

Dependent variable (Y)

Predictor (X)

Moderator (M)

Generated R code

Answer option

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GLM / binary variable

GLM / binary variable

Use sample

Dataset name

Binary response column

Value representing 1

Predictors, separated by commas

Generated R code

Answer option

Copy code

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

Interpret GLM output

Use sample output | Answer option | Copy all prose

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RStudio output

Paste the Call, Coefficients, and model summary from summary(glm(...)) here…

Predictor interpretations

HAART_A Copy

HAART_B Copy

VLoad Copy

Age Copy

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Confusion matrix

Confusion matrix

Use sample

Dataset name

Outcome column (0/1)

Fitted model object

Classification threshold

Generated R code

Answer option

Copy code

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

Interpret confusion matrix

Use sample output

Answer option

Copy all prose

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

RStudio output

Paste the confusion matrix and reported statistics here…

Generated interpretation

Classification counts i

Accuracy i

Sensitivity i

Specificity i

Precision i

Search a test, topic, or R function…

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PCA & component significance

PCA setup

Use sample

Dataset name

PCA variables, separated by commas

column1, column2, column3, colum

Components to retain

Downstream model

PCA workflow

Answer option

Copy all

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

1. Select PCA data Copy

2. Run PCA Copy

3. View summary Copy

4. View PC1 and PC2 loadings Copy

5. Add PC scores Copy

6. Draw scree plot Copy

7. Draw biplot Copy

Variance explained by retained components

Use sample output | Answer option | Copy prose

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

RStudio output

Paste the Importance of components table from summary(pca_model) here…

Generated interpretation

Individual | Cumulative

Component loading equations

Use sample output | Answer option | Copy prose

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

RStudio output

Paste the PCA loading matrix from pca_model$rotation here…

Human-readable loadings

Downstream linear-model significance

Use sample output | Answer option | Copy all prose

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RStudio output

Paste the Call, Coefficients, and model summary from summary(lm(...)) here…

PC significance interpretations

PC1 Copy

PC2 Copy

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Forward/Backward Stepwise Selection

Stepwise-selection setup

Use sample

Selection direction Forward | Backward

Dataset name

Outcome variable column1

Columns to exclude ID

Full-model object name

Generated R code

Answer option

Copy code

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K-means clustering

K-means setup

Use sample

Dataset name

Features, separated by commas

column1, column2, column3

Maximum K for elbow plot

Chosen number of clusters

Random seed

Random starts (nstart)

Scale variables

Generated R code

Answer option

Copy code

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Calculate axis intervals

Off: basic plotting parameters and automatic axes. Applies to all plots.

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- **K-fold cross-validation**
- lpSolve / Sensitivity analysis
- Shapiro–Wilk
- Pearson correlation test

K-fold cross-validation

Cross-validation setup

Use sample

Dataset name

Dependent variable

column1

Predictors, separated by commas

column2, column3, column4

Number of folds (K)

Generated R code

Answer option

Copy code

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

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Search a test, topic, or R function…

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- PCA & component significance
- Forward/Backward Stepwise Selection
- K-means clustering
- K-fold cross-validation
- **lpSolve / Sensitivity analysis**
- Shapiro–Wilk
- Pearson correlation test

lpSolve / Sensitivity analysis

Optimisation model

Reset all | Use sample

Paste an optimisation model table or text — Use parser sample

Paste a Markdown/ASCII model table, objective, constraints, and variable declarations here…

Direction

Variable type

Decision-variable names

Objective coefficients

☐ **Compute sensitivity**

Constraints

Constraint name	15, 30, 40		200	–
Constraint name	15, 30, 40		200	–
Constraint name	15, 30, 40		200	–
Constraint name	15, 30, 40		200	–
Constraint name	15, 30, 40		200	–
Constraint name	15, 30, 40		200	–
Constraint name	15, 30, 40		200	–
Constraint name	15, 30, 40		200	–
Constraint name	15, 30, 40		200	–
Constraint name	15, 30, 40		200	–

Generated R code

Answer option | Copy code

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

Model table

Copy as ASCII table

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

Sensitivity results

Use sample

Paste lpSolve console output

Paste lp_sol$solution, lp_sol$objval, lp_sol$status, duals, and sensitivity ranges here…

Optimal solution
0, 4, 1

Objective value
2766.667

Duals: constraints, then variables
0.01, 0.02, 0, -0.1, 0, 0

Objective coefficient lower bounds
0.10, 0.30, 0.50

Objective coefficient upper bounds
0.25, 0.55, 0.80

RHS lower bounds
150, 45, 0

RHS upper bounds
250, 80, 10

Sensitivity interpretation

Answer option | Copy prose

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

Solution — Copy

Shadow Price — Copy

Reduced Cost — Copy

Slack — Copy

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Search a test, topic, or R function…

24 operations

Universal detector
Paste first, identify automatically

OPERATIONS

Plotting & tables
Summarize for stats
Skewness & kurtosis
Wide table format
Confidence / prediction intervals
One-sample mean / proportion
Two-sample mean / variance
>2 samples: ANOVA
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K-fold cross-validation
lpSolve / Sensitivity analysis
Shapiro–Wilk
Pearson correlation test

Shapiro–Wilk

RStudio input Use sample

```
## Shapiro-Wilk normality test
##
## data: dataset$column1
## W = 0.42617,
## p-value < 2.2e-16
```

Prose output Answer option

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

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- lpSolve / Sensitivity analysis
- Shapiro–Wilk
- **Pearson correlation test**

Pearson correlation test

RStudio input Use sample

```
## Pearson's product-moment correlation
##
## data: dataset$column1 and dataset$column2
## t = 0.46122, df = 209, p-value = 0.6451
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.1036276 0.1662399
## sample estimates:
## cor
## 0.03188728
```

Prose output Answer option

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

Untitled document
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Plotting & tables

Plot setup

Use sample

I would like to plot a … of

column1 against

column3 , titled

Grouped barplot of Amount against I . The dataset is called

(default is "dataset").

Axis transformations

X-axis transformation

Y-axis transformation

Generated R code

Answer option

Copy code

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

Calculate axis intervals

Off: basic plotting parameters and automatic axes. Applies to all plots.

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Plotting & tables

Plot setup

Use sample

I would like to plot a … of

column2, column3, column4 against

column1 , grouped by

column3 , titled

Grouped mean bar plot of Column3 a . The dataset is called

(default is "dataset").

Axis transformations

X-axis transformation

Y-axis transformation

Generated R code

Answer option

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Plotting & tables

Plot setup

Use sample

I would like to plot a [ ] of

[column1] , titled

[Frequency bar plot of Column1] . The dataset is called

[ ] (default is "dataset").

Axis transformations

X-axis transformation [ ]

Y-axis transformation [ ]

Generated R code

Answer option [ ]

Copy code

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Calculate axis intervals

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Plotting & tables

Plot setup

Use sample

I would like to plot a … of

column1 against

column3 , titled

Side-by-side bar plot of Column1 aga . The dataset is called

(default is "dataset").

Bar arrangement

Axis transformations

X-axis transformation

Y-axis transformation

Generated R code

Answer option

Copy code

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

Calculate axis intervals

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- Pearson correlation test

Plotting & tables

Plot setup

Use sample

I would like to plot a … of

column1 , titled

Histogram of Column1 . The dataset is called

(default is "dataset").

Histogram breaks

Axis transformations

X-axis transformation

Y-axis transformation

Generated R code

Answer option

Copy code

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Calculate axis intervals

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Plotting & tables

Plot setup

Use sample

I would like to plot a … of

column1 , titled

Pareto analysis of Column1 . The dataset is called

(default is "dataset").

Target contribution (%)

Axis transformations

X-axis transformation

Y-axis transformation

Generated R code

Answer option

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Calculate axis intervals

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Plotting & tables

Plot setup

Use sample

I would like to plot a … of

column2, column3, column4 against

column1 , grouped by

column3 , titled

Mean bar plot of Column2, Column3, . The dataset is called

(default is "dataset").

Bar arrangement

Histogram breaks

Target contribution (%)

Axis transformations

X-axis transformation

Y-axis transformation

Generated R code

Answer option

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Calculate axis intervals

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Confidence / prediction intervals

Interval setup

Use sample

I want to calculate [ ] for [ ]

with [ ]

Dataset name

Column / parameter name
column1

Confidence level (%)

Planning standard deviation
4

Desired margin of error
0.5

Calculated result

Copy prose | Answer option [ ] | Copy code

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

Interpret RStudio interval output

Use sample output | Answer option [ ] | Copy prose

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

RStudio output

[1] 246

Generated interpretation

Untitled document
v041126_1200h

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Confidence / prediction intervals

Interval setup

Use sample

I want to calculate [ ] for [ ]

with [ ]

Dataset name

Column / parameter name
column1

Confidence level (%)

Sample size
64

Sample proportion (optional)
0.64

Define the proportion by

Population items
transactions

Value representing success

Calculated result

Copy prose | Answer option [ ] | Copy code

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

Interpret RStudio interval output

Use sample output | Answer option [ ] | Copy prose

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

RStudio output

```
upper_limit lower_limit
[1,] 20.33 19.32
```

Generated interpretation

Untitled document
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Search a test, topic, or R function…

24 operations

Universal detector
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OPERATIONS

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- Pearson correlation test

Confidence / prediction intervals

Interval setup

Use sample

I want to calculate [ ] for [ ]

with [ ]

Dataset name

Column / parameter name
column1

Confidence level (%)

Expected proportion
0.64

Desired margin of error
0.5

Calculated result

Copy prose | Answer option [ ] | Copy code

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

Interpret RStudio interval output

Use sample output | Answer option [ ] | Copy prose

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

RStudio output

```
[1] 246
```

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- Pearson correlation test

Confidence / prediction intervals

Interval setup

Use sample

I want to calculate [ ] for [ ]

with [ ]

Dataset name

Column / parameter name
column1

Confidence level (%)

Sample mean (optional)
50

Sample standard deviation (optional)
4

Sample size (optional)
64

Sample proportion (optional)
0.64

Define the proportion by

Population items
transactions

Value representing success

Desired margin of error
0.5

Calculated result

Copy prose | Answer option | Copy code

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One-sample mean / proportion

Test setup

Use sample

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

I want to test for

Dataset name: dataset

Response column: column1

Hypothesized value: 0.5

Significance level

Population filter (optional): Product == "Book"

Exclude missing response values

Population description: book sales transactions

Success condition (optional): Amount > 50

Value representing success: yes

Answer option

Copy code

RStudio output

Use sample output

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

Paste the t.test() or prop.test() output here…

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One-sample mean / proportion

Test setup

Use sample

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

I want to test for

Dataset name: dataset

Response column: column1

Hypothesized value: 50

Significance level

Population filter (optional): Product == "Book"

Exclude missing response values

Population description: book sales transactions

Success condition (optional): Amount > 50

Value representing success: yes

Answer option

Copy code

RStudio output

Use sample output

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Two-sample mean / variance

Fisher's F test

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I want to test

Dataset name

Column 1 (response)

Column 2 (grouping)

Answer option

Copy code

RStudio output

Use sample output

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Paste the two-sample test output here…

Untitled document
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- Shapiro–Wilk
- Pearson correlation test

Two-sample mean / variance

Independent two-sample t-test

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

I want to test

Paired samples

Dataset name

Column 1 (response)

Column 2 (grouping)

Alternative hypothesis

Answer option

Copy code

RStudio output

Use sample output

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

Paste the two-sample test output here…

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PCA & component significance

PCA setup

Use sample

Dataset name

PCA variables, separated by commas
column1, column2, column3, colum

Components to retain

Downstream model

Dependent variable
column5

PCA workflow

Answer option

Copy all

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

1. Select PCA data Copy

2. Run PCA Copy

3. View summary Copy

4. View PC1 and PC2 loadings Copy

5. Add PC scores Copy

6. Draw scree plot Copy

7. Draw biplot Copy

8. Fit downstream model Copy

Variance explained by retained components

Use sample output | Answer option | Copy prose

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

RStudio output

Paste the Importance of components table from summary(pca_model) here…

Generated interpretation

Individual | Cumulative

Component loading equations

Use sample output | Answer option | Copy prose

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

RStudio output

Paste the PCA loading matrix from pca_model$rotation here…

Human-readable loadings

Downstream linear-model significance

Use sample output | Answer option | Copy all prose

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

RStudio output

Paste the Call, Coefficients, and model summary from summary(lm(...)) here…

PC significance interpretations

PC1 Copy

PC2 Copy

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Forward/Backward Stepwise Selection
K-means clustering
K-fold cross-validation
lpSolve / Sensitivity analysis
Shapiro–Wilk
Pearson correlation test

PCA & component significance

PCA setup

Use sample

Dataset name

PCA variables, separated by commas
column1, column2, column3, colum

Components to retain

Downstream model

Dependent variable
column5

Classification threshold

PCA workflow

Answer option

Copy all

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

1. Select PCA data Copy

2. Run PCA Copy

3. View summary Copy

4. View PC1 and PC2 loadings Copy

5. Add PC scores Copy

6. Draw scree plot Copy

7. Draw biplot Copy

8. Fit downstream model Copy

Variance explained by retained components

Use sample output | Answer option | Copy prose

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

RStudio output

Paste the Importance of components table from summary(pca_model) here…

Generated interpretation

Individual | Cumulative

Component loading equations

Use sample output | Answer option | Copy prose

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

RStudio output

Paste the PCA loading matrix from pca_model$rotation here…

Human-readable loadings

Downstream linear-model significance

Use sample output | Answer option | Copy all prose

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

RStudio output

Paste the Call, Coefficients, and model summary from summary(lm(...)) here…

PC significance interpretations

PC1 Copy

PC2 Copy

Search a test, topic, or R function…

24 operations

Universal detector
Paste first, identify automatically

OPERATIONS

- Plotting & tables
- Summarize for stats
- Skewness & kurtosis
- Wide table format
- Confidence / prediction intervals
- One-sample mean / proportion
- Two-sample mean / variance
- >2 samples: ANOVA
- One-variable linear regression
- Multi-variable linear regression
- Model significance
- Pairwise model comparison
- Dummy variables / reference groups
- Interaction terms
- GLM / binary variable
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lpSolve / Sensitivity analysis

Optimisation model

Reset all | Use sample

Paste an optimisation model table or text — Use parser sample

Paste a Markdown/ASCII model table, objective, constraints, and variable declarations here…

Direction

Variable type

Decision-variable names

Objective coefficients

Constraints

Constraint name	15, 30, 40		200	–
Constraint name	15, 30, 40		200	–
Constraint name	15, 30, 40		200	–
Constraint name	15, 30, 40		200	–
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Generated R code

Answer option — Copy code

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Model table

Copy as ASCII table

Copied text appears here. Click it, press Ctrl+A, then Ctrl+C.

Sensitivity results

Use sample

Paste lpSolve console output

Paste lp_sol$solution, lp_sol$objval, lp_sol$status, duals, and sensitivity ranges here…

Optimal solution: 0, 4, 1

Objective value: 2766.667

Duals: constraints, then variables: 0.01, 0.02, 0, -0.1, 0, 0

Objective coefficient lower bounds: 0.10, 0.30, 0.50

Objective coefficient upper bounds: 0.25, 0.55, 0.80

RHS lower bounds: 150, 45, 0

RHS upper bounds: 250, 80, 10

Sensitivity interpretation

Answer option — Copy prose

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lpSolve / Sensitivity analysis

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lpSolve / Sensitivity analysis

Optimisation model

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☐ **Compute sensitivity**

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Generated R code

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Sensitivity interpretation

Answer option | Copy prose

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Solution — Copy

Shadow Price — Copy

Reduced Cost — Copy

Slack — Copy